

RES. 6-012 Introduction to Probability - Lecture 11 & 12
Textbook Problem Sets - Continuous RVs - Section 4.1 & 4.2

• Problem Set - Section 4.1:

1) $X \rightarrow \text{uniform } [-1; 1]$ $Y = \sqrt{|X|}$

$$F_Y(y) = P(Y \leq y) = P(\sqrt{|X|} \leq y) = \int_{\{x | \sqrt{|x|} \leq y\}} f_X(x) dx$$

$$= P(|X| \leq y^2) = P(-y^2 \leq X \leq y^2) = P(0 \leq X \leq y^2) + P(-y^2 \leq X \leq 0)$$

$$= 2P(0 \leq X \leq y^2) = 2y^2 = y^2$$

$$f_Y(y) = \frac{dF_Y}{dy} \cdot 2y^2 = 4y. \quad \text{No.} \quad \times$$

$$f_Y(y) = \frac{dF}{dy} y^2 = 2y.$$

PDF of $-\ln|X|$ $Y = -\ln|X|$

$$F_Y(y) = P(Y \leq y) = P(-\ln|X| \leq y) = P(\ln|X| \geq -y) = P(|X| \geq e^{-y})$$

$$= P(-e^{-y} \leq X \leq +e^{-y}) = P(X \geq e^{-y})$$

$$= 2P(0 \leq X \leq e^{-y}) = 2 \left(\frac{1}{2} \right) (e^{-y}) = P(X \leq -e^{-y})$$

$$= e^{-y}$$

$$f_Y(y) = \frac{dF_Y}{dy}(y) = \frac{dF}{dy} e^{-y} = e^{-y} \quad \checkmark$$

$$F_Y(y) = P(\ln|X| \geq -y) = P(X \geq e^{-y}) + P(X \leq -e^{-y})$$

$$= \int_{e^{-y}}^1 f_X(x) dx + \int_{-1}^{-e^{-y}} f_X(x) dx$$

$$f_X(x) = \frac{1}{b-a} = \frac{1}{(1)-(-1)} = \frac{1}{2}$$

$\hookrightarrow$ Uniform!!

$$= \int_{e^{-y}}^1 \frac{1}{2} dx + \int_{-1}^{-e^{-y}} \frac{1}{2} dx$$

$$= \frac{1}{2} x \Big|_{e^{-y}}^1 + \frac{1}{2} x \Big|_{-1}^{-e^{-y}}$$

$$= \frac{1-e^{-y}}{2} + \frac{1-e^{-y}}{2}$$

$$= 1-e^{-y}$$

$$\hookrightarrow f_Y(y) = \frac{dF}{dy} (1-e^{-y}) = e^{-y} \quad \checkmark$$

3) PDF of $|X|^{1/3}$
 $Y = |X|^{1/3} = \sqrt[3]{|X|}$

$$F_Y(y) = P(\sqrt[3]{|X|} \leq y) = P(|X| \leq y^3) = P(-y^3 \leq X \leq y^3) \\ = F_X(y^3) - F_X(-y^3)$$

$$f_Y(y) = \frac{dF}{dy} (F_X(y^3) - F_X(-y^3)) = f_X(y^3) \cdot 3y^2 + f_X(-y^3) \cdot 3y^2 \\ = 3y^2 f_X(y^3) + 3y^2 f_X(-y^3) \quad (\text{for } y > 0)$$

PDF of $|X|^{1/4}$

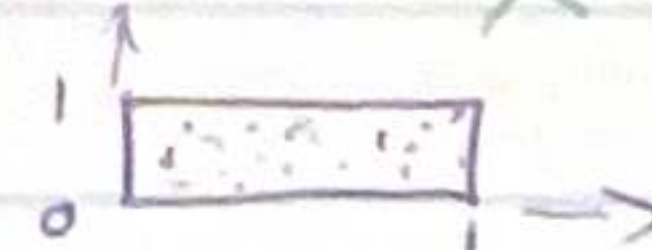
$$F_Y(y) = P(\sqrt[4]{|X|} \leq y) = P(|X| \leq y^4) = P(-y^4 \leq X \leq y^4) \\ = F_X(+y^4) - F_X(-y^4)$$

$$f_Y(y) = \frac{dF}{dy} (F_X(+y^4) - F_X(-y^4)) = 4y^3 f_X(y^4) + 4y^3 f_X(-y^4), \quad y > 0$$

5) CDF + PDF of $|X - Y|$; indep: $[0, 1]$

$Z = |X - Y|$

$$f_Z(z) = \int_0^1 f_X(x) f_Y(z-x) dx \\ = \int_0^1 (x)(-y) dx \\ = \frac{1}{2}(-y)x^2 \Big|_0^1 = -\frac{y}{2}(1) \\ = -\frac{y}{2}$$

→ 5) X & Y uniform $[0, 1]$ → 

$Z = |X - Y|$

$F_Z(|X - Y| \leq z) = P(-z \leq X - Y \leq z) =$

8) PDF $Z = X + Y$ X, Y indep, exponential common λ .

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

$$= \int_{-\infty}^{\infty} (\lambda e^{-\lambda x}) (\lambda e^{-\lambda(z-x)}) dx$$

$$f_Y(z-x) = \begin{cases} \lambda e^{-\lambda(z-x)} & z-x \geq 0 \\ 0 & z-x < 0 \end{cases}$$

$$z-x \geq 0 \Rightarrow x \leq z$$

$\hookrightarrow$ also $x \geq 0$ from exponential

$$\therefore f_Z(z) = \int_{-\infty}^z (\lambda e^{-\lambda x}) (\lambda e^{-\lambda(z-x)}) dx$$

$$= \lambda^2 \int_{-\infty}^z e^{-\lambda x - \lambda z + \lambda x} dx$$

$$= \lambda^2 \int_{-\infty}^z e^{-\lambda z} dx$$

$$= \lambda^2 (e^{-\lambda z}) \Big|_{-\infty}^z = 0$$

$$= \lambda^2 (1 - \cancel{e^{-\lambda z}})$$

$$\hookrightarrow f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 e^{-\lambda z} x \Big|_0^z = \lambda^2 z e^{-\lambda z} \checkmark$$

$\hookrightarrow$ Conditions

1. Exponential $f_X(x) \neq 0$ when $\underline{x \geq 0}$

2. $f_Y(z-x)$

$Y \rightarrow$ Exponential $f_Y(y) \neq 0$ when $\underline{y \geq 0}$

$$y \geq 0$$

$$\Rightarrow z-x \geq 0$$

$$x \leq z$$

Bounds f_Z $[0; z]$

12) PDF of $X + Y + Z$ uniform $[0, 1]$ independent.

$$K = X + Y$$

$$Z = x$$

$$Z = x + y$$

$$M = K + Z = X + Y + Z$$

$$M = K + Z$$

$$M - K$$

$$y \in [0, 1]$$

$$0 \leq x \leq 1$$

$$0 \leq x \leq 1$$

$$0 \leq y$$

$$0 \leq K - x$$

$$x \leq K$$

$$y \leq 1$$

$$K - x \leq 1$$

$$K \leq x + 1$$

$$x \geq K - 1$$

$$K - 1 \leq x \leq K$$

$$0 \leq x \leq 1$$

$$f_K(k) = \int_0^1 f_X(x) f_Y(k-x) dx = f_K(k)$$

$$f_K(k) = \int_0^k \left(\frac{1}{1-0}\right) \left(\frac{1}{k-0}\right) dx$$

$$= \int_0^k \frac{1}{k} dx =$$

$$f_K(k) = \int_0^k \left(\frac{1}{1-0}\right) \left(\frac{1}{1-0}\right) dx = \int_0^k 1 dx$$

$$= (k)_0^k$$

$$= k$$

$$f_M(m) = \int_0^m f_K(k) \cdot f_Z(m-k) dk$$

$$= \int_0^m (k) \cdot \left(\frac{1}{1-0}\right) dk$$

$$= \frac{1}{2} k^2 \Big|_0^m$$

$$= \frac{m^2}{2}$$

$$0 \leq Z \leq 1$$

$$0 \leq m - k \leq 1$$

$$0 \leq m - k$$

$$k \leq m \rightarrow m$$

$$m - k \leq 1$$

$$m \leq k + 1$$

$$k \leq m - 1$$

$\hookrightarrow f_K(k)$ for 2x uniforms $\rightarrow [0, 2]$

so two cases result

$$f_V(v) = v$$

$$0 \leq v \leq 1$$

$$f_V(v) = 2 - v$$

$$\underbrace{1 \leq v \leq 2}_{[0, 2]}$$

$\hookrightarrow$ Check resulting distributions.

12) ~~Work~~ Workthrough

$$X+Y+Z \quad \text{Let } V = X+Y \quad f_V \rightarrow [0, 2]$$

~~for~~ case: $[0, 1]$

$$\begin{aligned} f_V &= \int_0^K f_X(x) f_Y(v-x) dx \\ &= \int_0^K \left(\frac{1}{1-0}\right) \left(\frac{1}{1-0}\right) dx \\ &= V \end{aligned}$$

$$0 \leq y \leq 1-x$$

$$\begin{aligned} y &\leq 1 \\ v-x &\leq 1 \\ x &\leq v-1 \end{aligned}$$

case: $[1, 2]$

$$f_V = \int_{v-1}^1 f_X(x) f_Y(v-x) dx$$

$$1 < v < 2$$

$$x \leq v-1$$

$$x \leq v$$

$$= \int_{v-1}^1 \left(\frac{1}{1-0}\right) \left(\frac{1}{1-0}\right) dx$$

$$\begin{aligned} \max(0, v-1) \\ \min(1, v) \end{aligned}$$

$$= v - (v-1)$$

$$= 1 - (v-1)$$

$$= 2 - v$$

$$f_V = \begin{cases} v & ; 0 \leq v \leq 1 \\ 2-v & ; 1 \leq v \leq 2 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\text{Let } W = X+Y+Z = V+Z \quad [0, 3]$$

$$f_W(w) = \int f_V(v) f_Z(w-v) dv$$

$$f_V = 0 \text{ in } 0 \leq v \leq 2$$

$$f_W = 0 \text{ in } 0 \leq w-v \leq 1$$

case $w \leq 1$:

$$\begin{aligned} f_W(w) &= \int_0^w f_V(v) f_Z(w-v) dv = \int_0^w v dv \\ &= \frac{w^2}{2} \end{aligned}$$

$$\begin{aligned} 0 &\leq v \leq 2 \\ w-1 &\leq v \leq w \end{aligned}$$

case $1 \leq w \leq 2$:

$$f_W(w) = \int_{w-1}^w f_V(v) f_Z(w-v) dv$$

$$= \int_{w-1}^1 v dv + \int_1^w (2-v) dv$$

$$= \frac{1}{2} - \frac{(w-1)^2}{2} - \frac{(w-2)^2}{2} + \frac{1}{2}$$

$$= 1 - \frac{(w-1)^2}{2} - \frac{(w-2)^2}{2}$$

case $2 \leq w \leq 3$

$$f_w(w) = \int_{w-1}^2 f_v(v) f_z(w-v) dv = \int_{w-1}^2 (2-v) dv$$

$$= \frac{(3-w)^2}{2}$$

$$f_w(w) = \begin{cases} \frac{w^2}{2} & ; 0 \leq w \leq 1 \\ 1 - \frac{(w-1)^2}{2} - \frac{(w-2)^2}{2} & ; 1 \leq w \leq 2 \\ \frac{(3-w)^2}{2} & ; 2 \leq w \leq 3 \\ 0 & ; \text{otherwise.} \end{cases}$$

14) Light bulb lifetimes {independent} {exponential}

$$Z = \min\{X, Y\}$$

$$f_X = \lambda e^{-\lambda x} \quad \mu = 1/\lambda$$

$$1/X \sim 1/Y \sim \dots$$

$$\begin{aligned} X - Y &\geq 0 \\ X - Y &\leq 0 \\ Y - X &\leq 0 \end{aligned}$$

$$P(Z > z) = P(X > z \text{ and } Y > z)$$

$$= P(X > z) \cdot P(Y > z)$$

$$F_Z(y) = 1 - P(Z > z) = 1 - P(X > z) P(Y > z)$$

$$= 1 - \int_0^\infty \lambda e^{-\lambda z} dz \int_0^\infty \lambda e^{-\mu z} dz$$

$$= 1 - \lambda^2 \left[\frac{e^{-\lambda z}}{-\lambda} \right]_0^\infty \left[\frac{e^{-\mu z}}{-\mu} \right]_0^\infty = 1 - \frac{1}{\lambda^2} \cdot \frac{1}{\mu^2}$$

$$= 1 - \frac{1}{\lambda^2} (0 - 1)^2 = 1 - \frac{1}{\lambda^2}$$

$$\left\{ \begin{aligned} &= 1 - \int_0^\infty (e^{-\lambda z} \times e^{-\mu z}) dz \\ &= 1 - \int_0^\infty e^{-z(\lambda+\mu)} dz \end{aligned} \right\}$$

Problem Set - Section 4.2

$$\begin{aligned} 17) \quad \text{var}(X+Y) &= \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y) \\ &= \cancel{\text{var}(X)} \\ \text{var} \quad \text{var}(X) &= \text{var}(Y) \end{aligned}$$

$$\begin{aligned} \circ X+Y \Rightarrow \quad \text{var}(X+Y) &= \text{var}(X) + \text{var}(X) + 2\text{cov}(X, Y) \\ &= 2\text{var}(X) + 2\text{cov}(X, Y) \end{aligned}$$

$$\begin{aligned} \circ X-Y \Rightarrow \quad \text{var}(X-Y) &= \text{var}(X) + (-\text{var}(X)) + 2\text{cov}(X, Y) \\ &= 2\text{cov}(X, Y) \end{aligned}$$

$$2\text{var}(X) + 2\text{cov}(X, Y) \neq 2\text{cov}(X, Y) \quad \times$$

$$\hookrightarrow \text{cov}(X, Y) = E[XY] - \cancel{E[X]E[Y]}$$

$$\cancel{\text{cov}(X+Y, X-Y)} = E[(X+Y)(X-Y)] - \cancel{E[X+Y]E[X-Y]}$$

$\rightarrow$ Assume X and Y have zero mean

$\hookrightarrow$ Does not change spread of $\text{var}(X)$ or $\text{var}(Y)$

$\hookrightarrow$ Does not affect $\text{cov}(X, Y)$ to shift mean by a constant.

$\hookrightarrow$ Zero mean retains generality!!

Rule: $\text{cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$

$$\begin{aligned} \text{cov}(X+Y, X-Y) &= E[(X+Y - E[X+Y])(X-Y - E[X-Y])] \\ &= E[(X+Y)(X-Y)] \\ &= E[X^2 - Y^2] \\ &= E[X^2] - E[Y^2] \quad (1) \end{aligned}$$

$$\text{var}(X) = E[X^2] - (E[X])^2 = E[X^2]$$

$$\text{var}(Y) = E[Y^2] - (E[Y])^2 = E[Y^2]$$

$$\begin{aligned} \circ (1) \rightarrow \text{cov}(X+Y, X-Y) &= E[X^2] - E[Y^2] = \text{var}(X) - \text{var}(Y) \\ &= \text{var}(X) - \text{var}(X) \\ &= 0 \quad \circ \text{No correlation} \end{aligned}$$

18) (W, X) & (Y, Z) pairwise uncorrelated.

$$\begin{aligned} R &= W+X \\ S &= X+Y \\ T &= Y+Z \end{aligned}$$

$$\begin{aligned} \rho(W, X) &= 0 = \frac{\text{cov}(W, X)}{\sqrt{1 \cdot 1}} \\ \text{cov}(W, X) &= 0 \\ \rho(Y, Z) &= 0 = \frac{\text{cov}(Y, Z)}{\sqrt{1 \cdot 1}} \\ \text{cov}(Y, Z) &= 0. \end{aligned}$$

$$\text{cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y)$$

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$

$\rho(R, S)$:

$$\rho(R, S) = \frac{\text{cov}(R, S)}{\sqrt{\text{var}(R)\text{var}(S)}} = \frac{\text{cov}(W+X, X+Y)}{\sqrt{\text{var}(W+X)\text{var}(X+Y)}}$$

$$\begin{aligned} \text{cov}(R, S) &= \text{cov}(W+X, X+Y) = E[(W+X - E[W+X])(X+Y - E[X+Y])] \\ &= E[(0+0-0)(0+0-0)] \\ &= 0 \end{aligned}$$

$$\text{cov}(R, S) = E[RS] - E[R]E[S]$$

$$= E[(W+X)(X+Y)] - E[W+X]E[X+Y]$$

$$= E[WX + WY + X^2 + XY]$$

$$= E[X^2]$$

$$= \text{var}(X) = 1$$

$$\left\{ \begin{aligned} \text{var}(X) &= E[X^2] - (E[X])^2 \\ &= 1 - 0^2 = 1 \end{aligned} \right.$$

$$\text{var}(R) = \text{var}(W+X) = \text{var}(X) + \text{var}(W) + 2\text{cov}(W, X)$$

$$= (1) + (1) + 2(0) = 2$$

$$\text{var}(S) = \dots = 2$$

$$\therefore \rho(R, S) = \frac{\text{cov}(R, S)}{\sqrt{\text{var}(R)\text{var}(S)}} = \frac{0}{\sqrt{(2)(2)}} = 0$$

$\rho(R, T)$:

$$\text{cov}(R, T) = E[RT] - E[R]E[T] = E[(W+X)(Y+Z)]$$

$$= E[WY + ZW + XY + XZ] = 0$$

$$\text{var}(R) = \text{var}(W+X) = 2$$

$$\text{var}(T) = \text{var}(Y+Z) = \text{var}(Y) + \text{var}(Z) + 2\text{cov}(Y, Z) = (1) + (1) + 2(0) = 2$$

$$\therefore \rho(R, T) = \frac{\text{cov}(R, T)}{\sqrt{\text{var}(R)\text{var}(T)}} = \frac{0}{\sqrt{(2)(2)}} = 0$$

$$19) E[X] = 0 \quad E[X^2] = 1 \quad E[X^3] = 0 \quad E[X^4] = 3.$$

$$Y = a + bX + cX^2$$

$$\rho(X, Y) =$$

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}}$$

$$\begin{aligned} \text{cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= E[X(a + bX + cX^2)] \\ &= aE[X] + bE[X^2] + cE[X^3] \\ &= a(0) + b(1) + c(0) = b \end{aligned}$$

$$\text{var}(X) = E[X^2] - (E[X])^2 = (1) + (0)^2 = 1$$

$$\begin{aligned} \text{var}(Y) &= \text{var}(a + bX + cX^2) = E[(a + bX + cX^2)^2] - E[a + bX + cX^2]^2 \\ &= E[a^2 + 2abX + 2acX^2 + b^2X^2 + 2bcX^3 + c^2X^4] - E[\dots] \\ &= E[a^2 + (2ac + b^2)X^2 + c^2X^4] - E[\dots] \\ &= E[a^2 + (2ac + b^2)X^2 + c^2X^4] - (E[a + cX^2])^2 \end{aligned}$$

$$= a^2 + (2ac + b^2)(1) + c^2(1) - (a + c(1))^2$$

$$= a^2 + (2ac + b^2) + c^2 - a^2 - 2ac - c^2$$

$$= b^2$$

$$L.D = E[a^2 + (2ac + b^2)X^2 + c^2X^4] + \dots$$

$$= (a^2 + 2ac + b^2 + 3c^2) - (a^2 + c^2 + 2ac)$$

$$= b^2 + 2c^2$$

$$\rho(X, Y) = \frac{b}{\sqrt{a^2 + b^2 - a^2}}$$

$$\rho(X, Y) = \frac{(b)}{\sqrt{b^2 + 2c^2}}$$